

LOW STYRENE SBR: PATENT LANDSCAPE AND PROCESS ANALYSIS

1. ABSTRACT

This report analyzes patent evidence for low styrene styrene-butadiene rubber (SBR), defined here as SBR with reduced combined styrene content relative to conventional grades and, in the primary patent, explicitly 1-15 wt% combined styrene. The patent landscape in the provided manifest is narrow, with one primary disclosure focused on emulsion polymerization of low-styrene SBR using a mixed emulsifier system. The disclosed approach addresses a known tradeoff in low-styrene SBR: lower styrene content can improve compatibility and certain tire-bead-filler properties, but often reduces polymerization rate and can worsen thermal/mixing stability after compounding. The patent proposes a specific emulsifier combination of an aliphatic organic acid in acid form and a sulfonate-based compound, enabling low-styrene SBR latex formation at low temperature while maintaining acceptable polymerization rate and low coagulation.

The key synthesis route is emulsion polymerization of 1,3-butadiene and styrene, with Example 1 using 94 parts by weight butadiene and 6 parts by weight styrene, 10°C polymerization temperature, 0.05 parts by weight initiator, 0.5 parts by weight molecular weight regulator, and 200 parts by weight water per 100 parts monomer. Coagulation is performed with calcium chloride at 70°C, followed by aging, washing, filtration, and roll drying. Comparative examples vary the emulsifier composition, showing that mixed emulsifier systems outperform single-component systems in balancing polymerization time and coagulation. The patent therefore contributes a practical formulation/process strategy for producing low-styrene SBR suitable for blending with higher-styrene rubber in tire bead filler and related applications.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP3243844B1
- Patent Title: Method for preparing styrene-butadiene rubber having low combined styrene ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2022
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses low-temperature emulsion polymerization of 1,3-butadiene and styrene to make low styrene-butadiene rubber latex, followed by calcium-chloride coagulation, washing, filtration, and roll drying. The key process feature is use of a mixed emulsifier system comprising an aliphatic organic acid in acid form and a sulfonate-based compound to improve polymerization rate and reduce coagulation in low-styrene SBR production. - Combined styrene content: 1 wt% to 15 wt% — Claims - Combined styrene content: 5 wt% to 10 wt% — Claim 2 - Sulfonate-based compound content: 0.1 parts by weight to 5 parts by weight per 100 parts by weight monomer — Claims - Aliphatic organic acid content: 0.5 parts by weight to 7 parts by weight per 100 parts by weight monomer — Original claim text - Aliphatic organic acid carbon chain length: 12 to 18 carbon atoms — Claim 3 - Aliphatic organic acid examples: oleic acid, lauric acid, myristic acid, palmitic acid, stearic acid, eicosanoic acid — Claim 4 - Polymerization temperature: 10°C — Example 1 - 1,3-butadiene feed: 94 parts by weight — Example 1 - Styrene feed: 6 parts by weight — Example 1 - Aliphatic organic acid: 3.5 parts by weight — Example 1 - SANS: 0.5 parts by weight — Example 1 - Initiator: 0.05 parts by weight — Example 1 - Molecular weight regulator: 0.5 parts by weight — Example 1 - Water: 200 parts by weight per 100 parts by weight monomer — Example 1 - Conversion ratio at termination: 80% — Example 1 - Calcium chloride: 2 parts by weight per 100 parts by weight latex solids — Example 1 - Coagulation temperature: 70°C — Example 1 - Aging time: 20 minutes — Example 1 - Washing cycles: 2 to 3 times with ion exchange water — Example 1 - Example 2 aliphatic organic acid: 1.5 parts by weight — Example 2 - Example 2 SANS: 1.0 parts by weight — Example 2 - Example 3 aliphatic organic acid: 2.0 parts by weight — Example 3 - Example 3 SANS: 2.0 parts by weight — Example 3 - Comparative Example 1 aliphatic organic acid: 4.0 parts by weight — Comparative Example 1 - Comparative Example 2 SANS: 4.0 parts by weight — Comparative Example 2 - Comparative Example 3 rosin acid: 4.0 parts by weight — Comparative Example 3 - Comparative Example 4 aliphatic organic acid: 2.0 parts by weight — Comparative Example 4 - Comparative Example 4 rosin acid: 2.0 parts by weight — Comparative Example 4 - Comparative Example 5 SANS: 2.0 parts by weight — Comparative Example 5 - Comparative Example 5 rosin acid: 2.0 parts by weight — Comparative Example 5 - Example 1 polymerization time: 7.1 hr — Table 1 - Example 1 produced coagulation: 0.02% by weight — Table 1 - Example 2 polymerization time: 7.9 hr — Table 1 - Example 2 produced coagulation: 0.01% by weight — Table 1 - Example 3 polymerization time: 7.5 hr — Table 1 - Example 3 produced coagulation: 0.005% by weight — Table 1 - Comparative Example 1 polymerization time: 7.5 hr — Table 1 - Comparative Example 1 produced coagulation: 0.3% by weight — Table 1 - Comparative Example 2 polymerization time: 14 hr — Table 1 - Comparative Example 2 produced coagulation: 0.005% by weight — Table 1 - Comparative Example 3 polymerization time: 9.0 hr — Table 1 - Comparative Example 3 produced coagulation: 0.5% by weight — Table 1 - Comparative Example 4 polymerization time: 8.2 hr — Table 1 - Comparative Example 4 produced coagulation: 0.4% by weight — Table 1 - High styrene-butadiene rubber combined styrene content: 85% — Test Example 2 - High styrene-butadiene rubber particle size: 100 nm to 200 nm — Test Example 2 - Blend ratio of low styrene SBR to high styrene SBR: 2:8 — Test Example 2 - Milling temperature: 160°C — Test Example 2 - T5 measurement temperature/time basis: 160°C, 3 minutes — Test Example 2 -

Vulcanization temperature/time: 145°C for 45 minutes — Test Example 2

Relevant Experimental Evidence - Example 1: Demonstrated successful low-styrene SBR latex preparation at 10°C using 94/6 butadiene/styrene feed and mixed emulsifier; polymerization time was 7.1 hr with 0.02% coagulation. - Example 2: Showed that lowering aliphatic organic acid and increasing SANS to 1.5/1.0 parts by weight still produced low coagulation (0.01%) with 7.9 hr polymerization time. - Example 3: Showed the best coagulation suppression among the examples, with 0.005% coagulation and 7.5 hr polymerization time using 2.0/2.0 parts by weight emulsifier split. - Comparative Example 2: Using SANS alone gave the longest polymerization time, 14 hr, despite low coagulation, indicating poor rate performance without the acid/sulfonate combination. - Test Example 2: Low styrene SBR was blended with high styrene SBR (85% combined styrene) at 2:8 to evaluate mixture properties for formed rubber articles.

Technical Relevance This patent is directly relevant because it explicitly targets low styrene SBR and defines a practical styrene-content window of 1-15 wt% combined styrene. It provides a concrete emulsion-polymerization recipe and emulsifier strategy for improving polymerization rate and stability, which are central challenges in low-styrene SBR manufacture.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The evidence supports emulsion polymerization with controlled low styrene feed as the most directly disclosed route to low styrene SBR. The strongest parameter set in the provided patent uses 94 parts by weight 1,3-butadiene and 6 parts by weight styrene, 10°C polymerization, and a mixed emulsifier of aliphatic organic acid plus sulfonate-based compound, with coagulation by calcium chloride at 70°C. The patent indicates that balanced emulsifier ratios reduce coagulation and preserve polymerization rate better than single-emulsifier systems. For low-styrene SBR synthesis, the recommended evidence-based target is 1-15 wt% combined styrene, preferably 5-10 wt%, with low-temperature emulsion polymerization and mixed emulsifier control to maintain processability and downstream blending performance.

5. REFERENCES

- EP3243844B1 | Method for preparing styrene-butadiene rubber having low combined styrene ... | Not disclosed | EP | 2022 | <https://patents.google.com/patent/EP3243844B1/en>